Functional Description of Diagnosis

DFPV S15

ABS9/ESP9

Protocol: UDS on CAN

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1 History

All corrections of misspelling are excluded from the following list.

Ver.	Date	Editor	Chapter	Description of changes
01.00	10.22.2012	Shao Weifeng		Initial Version Based On UDS ISO14229 and Diaf of DFLZ B20/C20 Projects
			5.3	CAN ID Modification: Physical Addressing Request Messages →7C2 Physical Addressing Request Messages →7CA Functional Addressing Request Messages →7DF
			8.1	DTC Code change: 0x500004-->0x502004 0x501008-->0x503008
			6.9.3.2 6.9.3.5	Add Support for request result of SAS calibration

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3 Referenced Documents

3.1 International and National Documents

The general requirements of CAN diagnosis communication protocol are based on Unified Diagnostic Services (UDS) Design Standard requirements, specified in the following documents:

1. ISO 14229-1: Road Vehicles – Diagnostic Systems: Unified Diagnostic Services(UDS)
2. ISO 15765-2: Road Vehicles – Diagnostics on CAN: Network Layer Services
3. ISO 15765-3: Road Vehicles – Diagnostics on CAN: Application Layer Services
4. ISO 15031-6: Road Vehicles – Communication between vehicle and external equipment for emissions-related diagnostics — Part 6 DTC definitions

Remark:
Requirements in the above referenced documents may not be fully implemented. The document Functional Description of Diagnosis is the agreed specification for the implemented customer diagnosis.

3.2 Customer Documents

The customer specific requirements are based on the following diagnosis related documents:

- UDS on CAN Implementation Requirements

Remark:
Requirements in the above referenced documents may not be fully implemented. The document Functional Description of Diagnosis is the agreed specification for the implemented customer diagnosis.

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4 Introduction

CAN diagnosis provides the capability of quickly accessing the information and functions of the control unit over the **C**ontroller **A**rea **N**etwork (CAN), without disconnecting the cable harness. CAN diagnosis provides services to:

- Identify the ECU
- Read input signals of the ECU
- Write output signals to the actuators
- Read ECU-internal information, such as the contents of the fault memory

CAN diagnosis can be used for end-of-line testing or in the service workshop. The testing time can be reduced substantially by using the services provided by CAN diagnosis.

This document describes the diagnosis functions and specifications of the Default, Extended (EOL) and customer specific Diagnostic Session (Application Software).

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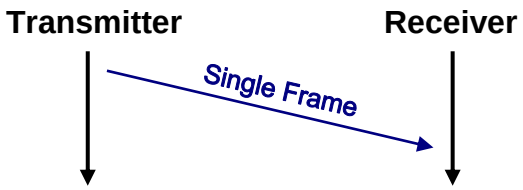
5 Communication on CAN

5.1 Data Stream

The ECU supports CAN diagnostic messages using the Unacknowledged Segmented Data Transfer (USDT) mechanism as specified in ISO 15765-2. Accordingly the following two types of CAN data streams are supported:

5.1.1 Non-segmented data stream

Consists of a Single Frame (SF) of non-segmented CAN message. This can be used for messages with maximum data length of 7 data bytes.



5.1.2 Segmented data stream

A segmented data stream consists of multiple frames of segmented CAN messages. A segmented data stream must be used if the maximum data length exceeds 7 data bytes in case of normal addressing.

The data stream comprises a First Frame (FF) and a number of Consecutive Frames (CF) containing the subsequent data bytes, following the First Frame. The receiver has to send a Flow Control (FC) after reception of the First Frame.

The Flow Control contains information regarding Block Size and the Separation Time. The Block Size (BS) is the maximum number of Consecutive Frames the receiver allows the sender to send, before waiting for an authorisation (Additional Flow Control Frames) from the receiver to continue transmission. The Separation Time (Stmin) is the (minimum) time the sender has to wait between the transmissions of two consecutive frames.

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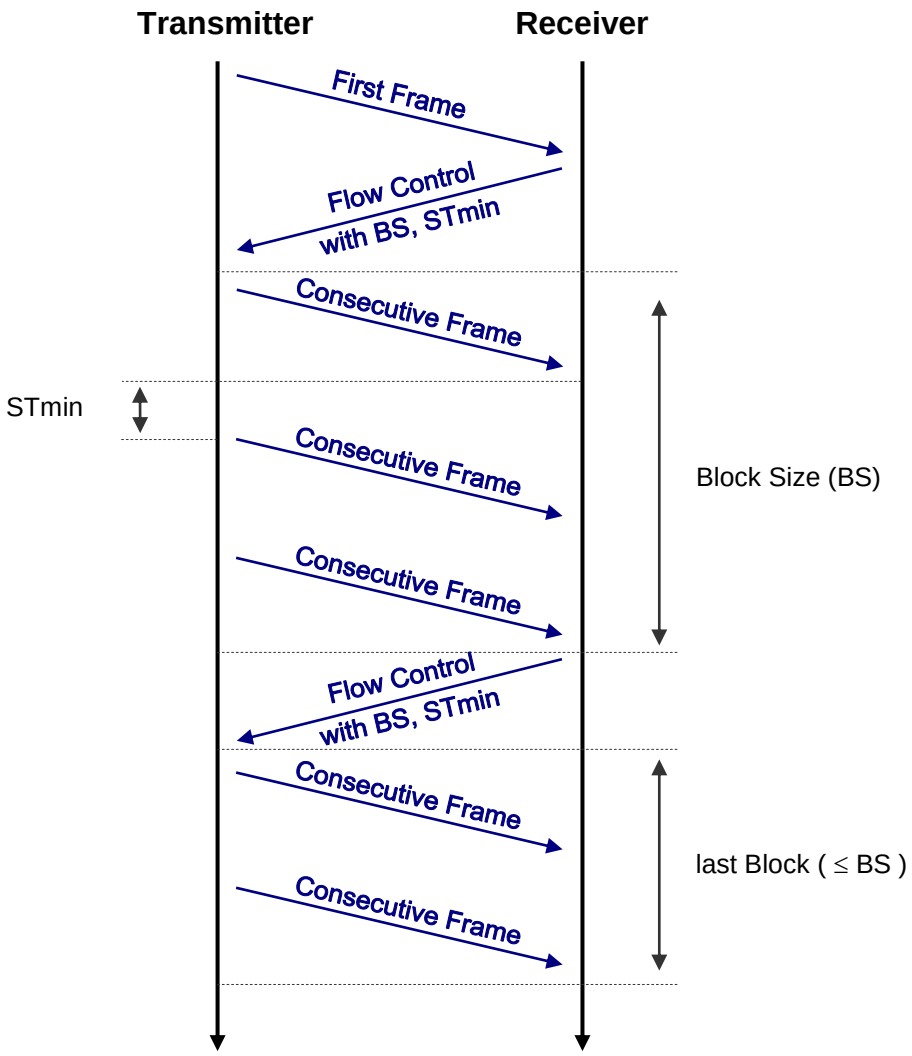

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5.2 Message Types

Each Single Frame (SF), First Frame (FF), Consecutive Frame (CF) or Flow Control (FC) has an 8-byte Data Field (DLC of CAN message is set to 8) consisting of the following parts:

- Protocol Control Information (PCI)
  indicates the frame type, number of frames, data length etc.
- Transmission data (Data)
  contains data to be sent

Data Field Structure					
Frame Type	**Byte 1**		**Byte 2**	**Byte 3**	**Byte 4 – 8**
	Bits 7-4	**Bits 3-0**			
Single Frame (SF)	0000	SF_DL	DB 1	DB 2	DB 3 – 7
First Frame (FF)	0001	FF_DL		DB 1	DB 2 – 6
Consecutive Frame (CF)	0010	SN	DB x_1	DB x_2	DB x_3-x_7
Flow Control (FC)	0011	FS	BS	Stmin	N/A

SF_DL: 4 Bit Single Frame Data length
SN: Sequence Number
BS: Block Size
DB: Data byte
FF_DL: 12 Bit First Frame Data length
FS: Status of data stream
Stmin: Minimum Separation Time

5.2.1 Single Frame

Single Frame message type is used for data blocks with a maximum data length of 7 Bytes and is identified by the upper nibble of PCI set to 0000b. Single Frame Data length SF_DL is computed as:

Single Frame Data Length (SF_DL):

Hex Value	Description
0h – 7h	Single Frame Data length (SF_DL) contains number of data Bytes

5.2.2 First Frame

First Frame message type is used for data blocks more than 7 Bytes long, in combination with one or more Consecutive Frames and is identified by the upper nibble of PCI set to 0001B. First Frame Data length FF_DL is computed as:

First Frame Data Length (FF_DL):

Hex Value	Description of FF_DL
000h – 007h	Reserved
008h – FFFh	First Frame Data length (FF_DL) contains number of data bytes. 12 Bit FF_DL theoretically supports sending of up to 4096 data bytes. However the current implementation does not need to support this maximum of 4096 data bytes.

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5.2.3 Consecutive Frame

Consecutive Frames follow First Frame and is identified by the upper nibble of PCI set to 0010B. The lower nibble of the PCI has a Sequence Number (SN) which is determined as follows:

- First Frame is regarded as containing Sequence Number 0.
- Hence the first Consecutive Frame always holds Sequence Number 1.
- Sequence Number is incremented by one for every additional Consecutive Frame.
- A Flow Control frame sent “in between” does not affect the Sequence Number counting.
- When the Sequence Number reaches the value of 15, the following Consecutive Frame will be sent with Sequence Number set to 0.

Definition of Sequence Number (SN):

Protocol Data name	FF	CF	CF	CF	CF	CF	CF	CF	CF
SN (hex)	0h	1h	2h	…	Eh	Fh	0h	1h	…

Hex Value	Description
0h – Fh	Sequence Number (SN) The Sequence Number has a range of 0 to 15. SN-Value is stored to the lower 4 bits of the Byte 1.

5.2.4 Flow Control

After the First Frame or the last Consecutive Frame of a block of frames when further CFs are required to complete the data transfer, a Flow Control message must be sent by the receiver node, and is identified by the upper nibble of PCI set to 0011b. With the Flow Control, the receiving node notifies the transmitting node how many messages it can receive between two Flow Control frames (maximum Block Size, Bsmax) and at what rate between Consecutive Frames (minimum Separation Time, Stmin). The lower nibble of the Flow Control PCI shall have a Flow Control Status (FS) set to “Continue To Send (CTS)”.

Stream Status (FS):

Hex Value	Description
0h	Continue To Send (CTS) The Receiver is ready and awaits the BS announced maximal amount of Consecutive Frames
1h	Wait (WT) The FlowControl Wait parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to “1”. It shall cause the sender to continue to wait for a new FlowControl N_PDU and to restart its N_BS timer.
2h	Overflow (OVFLW) The FlowControl Overflow parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to “2”. It shall cause the sender to abort the transmission of a segmented message and make an N_USData.confirm service call with the parameter <N_Result>=N_BUFFER_OVFLW. This N_PCI FlowStatus parameter value is only allowed to be transmitted in the FlowControl N_PDU that follows the FirstFrame N_PDU and shall only be used in case the message length FF_DL of the received FirstFrame N_PDU exceeds the buffer size of the receiving entity.

Remark: This parameter is configured to 0h in software.

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Block Size (Bsmax):

Hex Value	Description
00h	Block Size (Bsmax) No more Flow Controls will be send during the segmented Message. All Consecutive Frames will be send without a stop for an additional Flow Control to the Receiver.
01h – FFh	Block Size (Bsmax) Defines the maximal amount of Consecutive Frames till next Flow Control.

Remark: This parameter is configured to 00h in software.

Separation Time (Stmin):

Hex Value	Description
00h – FFh	Separation Time (Stmin) The Separation Time defines the minimal allowed time (in ms) between 2 Consecutive Frames.

5.3 Addressing Modes

5.3.1 Physical Addressing Mode

Physically addressing is used for a point –to-point communication between the diagnostic tool and the ECU (1:1 communication).
The diagnostic tool is sending a physically addressed request messages with CAN identifier CANID_REQ_PHYS to the ECU and the ECU will generally send a response message with CAN Identifier CAN_RES_PHYS.

Requirements

. Physically addressing (1:1 communication) is supported for all types of network layer messages.

5.3.2 Functional Addressing Mode

Functional addressing is used for broadcast communication between the diagnostic tool and all connected ECUs (1:n communication).
The diagnostic tool is sending a Functional addressed request messages with CAN identifier CANID_REQ_FUNC to all ECUs and each ECU will generally send a response message with CAN Identifier CAN_RES_PHYS.

Requirements

. Functional addressing (1:n communication) is only supported for Single Frame communication.
The ECU is ignoring all other functional addressed frames.

S15:

Addressing Type	Name	CAN-ID
CAN ID for physical addressing request messages	CANID_REQ_PHYS	7C2h
CAN ID for physical addressing response messages	CANID_RES_PHYS	7CAh
CAN ID for functional addressing request messages	CANID_REQ_FUNC	7DFh

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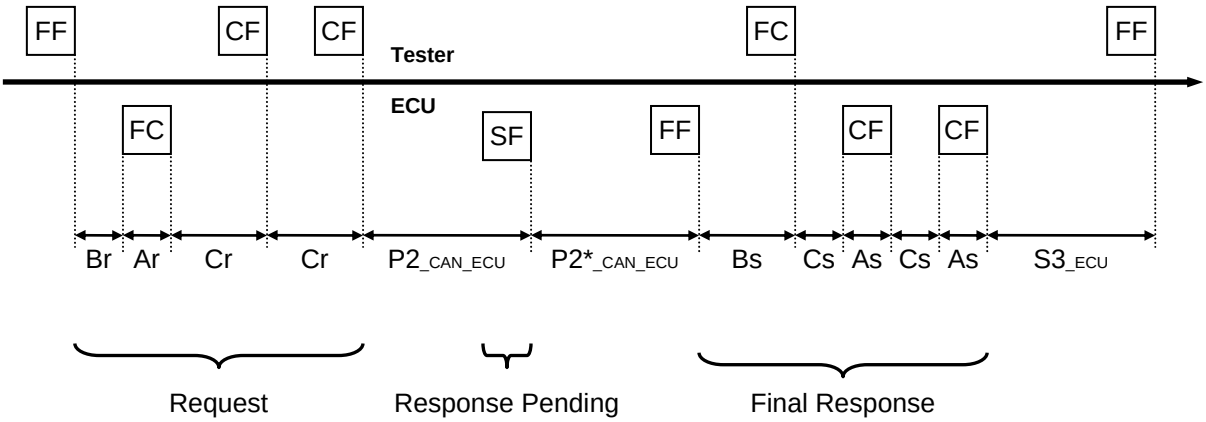
5.4 CAN Timing Parameters

Requirements:

- Communication speed (baud rate) – 500 kBd ±2 kBd
- ECU timings have an accuracy error < ± 2 ms



5.4.1 Performance Parameters

Parameter	Name	Min Value	Max Value
Performance requirement for the ECU to transmit a required Flow Control Frame after reception of a First Frame or Consecutive Frame from Tester.	Br	NA	25 ms
Performance requirement for the ECU to transmit a First Consecutive Frame after reception of a Flow Control. Performance requirement for the ECU to transmit a Next Consecutive Frame after sending of a Consecutive Frame.	Cs	N/A	100ms
Separation time –The Transmitter shall wait for more than Receiver Stmin. Value before sending the next Consecutive Frame.	Stmin	10ms	
Performance requirement for the ECU to start with the response message after the reception of a request message.	$P2_{CAN_ECU}$	N/A	50 ms
Performance requirement for the ECU to start with a further response message after transmission of a negative response message with negative response code 78h(Response Pending).	$P2_{CAN_ECU}$*	N/A	5000 ms

5.4.2 Timeout Parameters

Parameter	Name	Min Value	Max Value
Timeout for a ECU to access a network and to transmit a diagnostic CAN frame (time for transmission of the CAN frame on the sender side).	As/Ar	N/A	25 ms
Timeout for a ECU to wait for a Flow Control from the tester after sending First Frame (or a Next Consecutive Frame).	Bs	N/A	75 ms

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Time between reception of Flow Control and transmission of following Consecutive Frame or Time between transmission of two Consecutive Frames.	Cr	N/A	150ms
Timeout for a ECU wait for any diagnostic request message to keep a diagnostic session other than Default Session active.	S3_server	N/A	5000 ms

5.5 Communication Failures

5.5.1 Unexpected Messages

As a general rule, the arrival of an unexpected message or a message with illegal format from any node will be ignored.

5.5.2 Missing Messages

If the ECU senses a communication failure (e.g. timing violations of the Network Layer) during a tester request, the ECU discards the request and no response will be sent.

The tester has to monitor that no ECU-response has been sent within $\mathbf{P2}_{\text{CAN-ECU}}$, and hence the tester can repeat his previous request.

During Extended Diagnostic Session, if no request is received from the tester for $\mathbf{s3}_{\text{-server (MAX)}}$ (criteria for broken communication wire) ECU will exit to Default Session.

5.6 Condition and Behaviour of the ECU

5.6.1 Speed Limit

The ECU monitors a speed limit during the entire Extended Mode (EOL diagnosis mode). Speed limit is 10 Km/h.

5.6.2 Exit Conditions

The ECU will terminate an active Extended mode in the following cases:

- The speed limit is reached. Note that speed limit can be disabled by using the RoutineControl service
- Request for switching to Default Session is received .
- The ECU does not receive any tester message for more than 5 seconds (S3_server Timeout).

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6 Diagnostic Service List

ECU and Tester should support the diagnostic services listed below.

Diagnostic Service Name	Req. SID	Req. Sub Func	Supported Session		Addressing type		Security Access Level Required
			01	03	Phy	Fun	
DiagnosticSessionControl	10h				Y	Y	N
Default Session		01h	Y	Y			
Extended Session		03h	Y	Y			
ECUReset	11h		Y	Y	Y	Y	N
Software Reset		03h					
ClearDiagnosticInformation	14h		Y	Y	Y	Y	N
ReadDTCInformation	19h		Y	Y	Y	Y	N
ReportNumberOfDTC ByStatusMask		01h					
ReportDTCByStatusMask		02h					
ReportDTCSnapshotRecordBy DTCNumber		04h					
ReadDataByIdentifier	22h		Y	Y	Y	Y	N
WriteDataByIdentifier	2Eh		N	Y	Y	N	Y
SecurityAccess	27h						
RequestSeed		01h	N	Y	Y	N	N
SendKey		02h	N	Y	Y	N	N
CommunicationControl	28h		N	Y	Y	Y	N
EnableRxAndTx		00h					
DisableRxAndTx		03h					
RoutineControl	31h		N	Y	Y	N	Y
StartRoutine		01h					
StopRoutine		02h					
RequestRoutineResults		03h					
TesterPresent	3Eh		Y	Y	Y	Y	N
ControlDTCSetting	85h		N	Y	Y	Y	N
DTCstoringOn		01h					
DTCstoringOff		02h					

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6.1 DiagnosticSessionControl($10h)

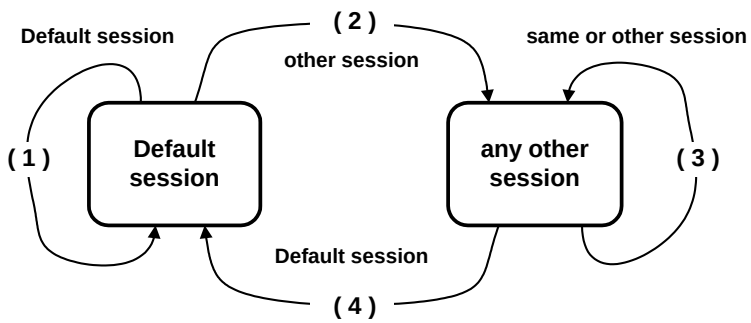
6.1.1 Description

The DiagnosticSessionControl service is used to enable different diagnostic sessions on the ECU. The ECU supports the following Diagnostic Sessions:

- **Default Session:**
  The ECU enables the Default session on ignition. In this session both the controller functions and diagnostic services are active. But this session does not support diagnostic services in Extended Diagnostic Session (see “Diagnostic Service List” section for services supported during this session).

- **Extended Session:**
  The Extended Session allows dedicated actuator activations and advanced Troubleshooting routines.
  As long as the Extended Session is running, no controller is active to prevent actuating conflicts. Additionally, this session is protected by a speed limit (10km/h). The session cannot be activated if the speed limit is exceeded. If the speed limit is exceeded a currently active Extended Session is terminated. For special situation, it is possible to deactivate the speed limit monitoring with **service $31.**

State Transitions



- The ECU is initializing the default session. The ECU is resetting all activated settings & controls. The Security Access state will be set to ECU “locked” (The ECU’s security feature is enabled).
- When the server transitions from any diagnostic session other than the Default session to the DefaultSession then the server shall reset security.

6.1.2 Request

Diagnostic Session Type
Select the desired diagnostic session. Refer to the subsections below for additional details.

PARAMETER: DiagnosticSessionType		
Size (Byte)	**Description**	**Value**
1 (Bit 0-6)	Default Session	01h

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PARAMETER: DiagnosticSessionType		
Size (Byte)	**Description**	**Value**
1 (Bit 0-6)	Extended Session	03h

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	10h
1	Sub Function Bit 0-6 Diagnostic Session Type Bit 7 Suppress Positive Response Message Indication Bit 0 = ECU is sending a positive response message 1 = ECU is not sending a positive response message	01/81h 03/83h

6.1.3 Positive response

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	50h
1	SubFunction	xxh
2,3	P2_CAN_ECU_max	xxxxh
4,5	P2* _CAN_ECU_max	xxxxh

Remark

P2_CAN_ECU_max 1LSB = 1ms

P2* _CAN_ECU_max 1LSB=10ms

6.1.4 Negative Response

ECU → TESTER: [SERVICE] – Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	10h
2	**Response Code** Sub Function Not Supported Incorrect Message Length Conditions Not Correct	 12h 13h 22h

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6.2 ECUReset ($11h)

6.2.1 Description

The service requests the ECU to effectively perform a software reset.

6.2.2 Request

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	11h
1	Sub Function Bit 0-6 Reset Type [03h] = Software Reset Bit 7 Suppress Positive Response Message Indication Bit [0] = ECU is sending a positive response message [1] = ECU is not sending a positive response message	03/83h

6.2.3 Positive response

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	51h
1	SubFunction	xxh

6.2.4 Negative Response

ECU → TESTER: [SERVICE] – Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	11h
2	**Response Code** Sub Function Not Supported Incorrect Message Length Response Pending General Reject	 12h 13h 78h 10h

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6.3 ClearDiagnosticInformation($14h)

6.3.1 Description

The service ClearDiagnosticInformation is used to clear the diagnostic information stored in the ECU memory.

6.3.2 Request

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	14h
1	GroupOfDTC – High byte	FFh
2	GroupOfDTC –Middle byte	FFh
3	GroupOfDTC –Low byte	FFh

6.3.3 Positive response

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	54h

6.3.4 Negative Response

ECU → TESTER: [SERVICE] – Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	14h
2	**Response Code** Incorrect Message Length Request Out Of Range Response Pending General Reject	 13h 31h 78h 10h

Remark: If the erasing of EEPROM needs more than $P2_{MAX}$, the ECU will send a negative response 78h (RequestCorrectlyReceived-ResponsePending) until it has been finished.

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6.4 ReadDTCInformation($19h)

6.4.1 Description

The service ReadDTCInformation allows a client to read DTC information from the ECU. The type of information which is read depends on the selected subfunction.

StatusOfDTC	
Position (Bit)	**Description**
0	testFailed
1	testFailedThisOperarionCycle
2	Reserved(0)
3	confirmedDTC
4	testNotCompletedSinceLastClear **(Not Support)**
5	testFailedSinceLastClear
6	testNotCompletedThisOperationCycle **(Not Support)**
7	WarningIndicatorRequested

Remark: The fault path is indicated by the stored DTC. When saving a new fault, a check is first conducted in order to establish whether the same DTC has already been saved at any point in the fault memory. If this is the case, the DTC is saved at the same position. Otherwise, it is saved at the next free position in the fault memory.

6.4.2 Request

The service supports following sub-functions.

PARAMETER: Subfunction		
Size (Byte)	**Description**	**Value**
1	ReportNumberOfDTCByStatusMask	01h
1	ReportDTCByStatusMask	02h
1	ReportDTCSnapshotRecordByDTCNumber	04h

Remark: When sending $19 with subfunction 0x02 & 0x01, setting DTCStatusMask to 0x08 (status bit ConfirmedDTC set to 1) will retrieve the stored DTC/Number of DTC in EEPROM (including Current and History Fault)

6.4.2.1 ReportNumberOfDTCByStatusMask

The service ReportNumberOfDTCByStatusMask can be used to call the ECU to transmit the number of diagnostic trouble codes matching a defined status mask to the diagnostic tool.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	19h
1	Subfunction ReportNumberOfDTCByStatusMask	01h
2	DTCStatusMask	xxh

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6.4.2.2 *ReportDTCByStatusMask*

The service ReadDTCByStatusMask can be used to call the ECU to transmit a list of diagnostic trouble codes and corresponding status information matching a defined status mask to the diagnostic tool.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	19h
1	Subfunction ReportDTCByStatusMask	02h
2	DTCStatusMask	xxh

6.4.2.3 *ReportDTCSnapshotRecordByDTCNumber*

The service ReadDTCSnapshotRecordByDTCNumber can be used to call the ECU to transmit Snapshot Record(s) matching a defined DTC to the diagnostic tool. This functionality is used to read environmental data from the fault memory.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	19h
1	Subfunction ReportDTCSnapshotRecordByDTCNumber	04h
2	DTCHighByte	xxh
3	DTCMiddleByte	xxh
4	DTCLowByte	xxh
5	DTCSnapshotRecordNumber	01h

6.4.3 Positive response

6.4.3.1 *ReportNumberOfDTCByStatusMask*

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	59h
1	Subfunction ReportDTCByStatusMask	01h
2	DTC StatusAvailablilityMask	ABh
3	DTC Format Identifier [00h] ISO15031-6 DTC Format	00h
4..5	DTC Count Byte #1 DTC Count High Byte Byte #2 DTC Count Low Byte	 xxh xxh

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6.4.3.2 *ReportDTCByStatusMask*

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	59h
1	Subfunction ReportDTCByStatusMask	02h
2	DTC StatusAvailablilityMask	ABh
	DTC #1 (Optional)	
3	Byte #1 DTC #1 High Byte	xxh
4	Byte #2 DTC #1 Middle Byte	xxh
5	Byte #3 DTC #1 Low Byte	xxh
6	StatusOfDTC #1 (Optional)	xxh
…	…	…
	DTC #n (Optional)	
4n-1	Byte #4n-1 DTC #1 High Byte	xxh
4n	Byte #4n DTC #1 Middle Byte	xxh
4n+1	Byte #4n+1 DTC #1 Low Byte	xxh
4n+2	StatusOfDTC #n (Optional)	xxh

Remark: The parameters DTC #1, #n and their Status are only present if at least one DTC is available to be reported according to the requested DTC Status Mask.
Remark: If there is no DTC status matching the requested DTC status, the ECU is sending a positive response with the parameter ABh.

6.4.3.3 *ReportDTCSnapshotRecordByDTCNumber*

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	59h
1	Subfunction ReportDTCByStatusMask	04h
2	DTCHighByte	xxh
3	DTCMiddleByte	xxh
4	DTCLowByte	xxh
5	StatusOfDTC	xxh
6	DTCExtendedDataRecordNumber	01h
7	DTCExtendedDataRecordNumberofIdentifiers	03h
8	ECU Voltage DataIdentifier(MSB)	FEh
9	ECU Voltage DataIdentifier(LSB)	00h
10	ECU Voltage (High Byte)	Xxh
11	ECU Voltage (Low Byte) (1LSB = 0.08V)	xxh
12	Vehicle Speed DataIdentifier (MSB)	FEh
13	Vehicle Speed DataIdentifier (LSB)	01h
14	Vehicle Speed (High Byte)	xxh
15	Vehicle Speed (Low Byte) (1LSB = 0.05625km/h)	xxh
16	Occurrence Counter DataIdentifier(MSB)	FEh
17	Occurrence Counter DataIdentifier(LSB)	02h
18	Occurrence Counter(High Byte)	Xxh
19	Occurrence Counter(Low Byte)	xxh

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Remark: If no DTC matches the requested DTC, the ECU is sending a negative response with the parameter 31h.

6.4.4 Negative response

ECU → TESTER: [SERVICE] – Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	19h
2	**Response Code** Sub Function Not Supported Incorrect Message Length Request Out Range Response Pending General Reject	 12h 13h 31h 78h 10h

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6.5 ReadDataByIdentifier ($22h)

6.5.1 Description

This service allows the diagnostic tool to request data record values from the ECU identified by Record Data Identifiers (DID). The client request message contains one 2-byte Data Identifier values that identifies the requested Data Record.

6.5.2 Request

PARAMETER: Data Identifier (DID)		
Size (Byte)	**Description**	**Value**
2	ActiveDiagnosticSession	F186h
2	VehicleManufacturerECUHardwareNumber – Customer PartNumber	F187h
2	Vehicle Identification Number -VIN	F190h
2	SoftwareVersionNumber	F195h
2	SpeedInformation	FD00h
2	InputData	FD01h
2	ActuationState	FD02h
2	FillingInStatus	FD03h
2	EOLStatus	FD04h
2	SystemSensor **(ESP only)**	FD05h
2	Variant Code **(ESP only)**	FD06h

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	22h
1..2	DID	xxxxh

6.5.3 Positive response

6.5.3.1 ActiveDiagnosticSession

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	62h
1..2	DID ActiveDiagnosticSession	F186h
3	Diagnostic Session Type 0x01: Default Session 0x03 : Extend Session	xxh

6.5.3.2 VehicleManufacturerECUHardwareNumber – Customer PartNumber

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	62h
1..2	VehicleManufacturerECUHardwareNumber – Customer PartNumber	F187h
3--16	Customer PartNumber, 14 BYTEs	xxh

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6.5.3.3 Vehicle Identification Number -VIN

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	62h
1..2	Vehicle Identification Number -VIN	F190h
3--19	VIN Code , 17 BYTEs	xxh

6.5.3.4 SoftwareVersionNumber

The identifier returns the SoftwareVersionNumber.

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	62h
1..2	DID SoftwareVersionNumber	F195h
3..10	SoftwareVersionNumber (ASCII, 8 bytes)	xx..xxh

6.5.3.5 SpeedInformation

The identifier allows reading information of the current vehicle and wheel speeds.

ECU → TESTER: [SERVICE] – Positive Response				
Byte	**Contents**	**Value + Unit**	**LSB**	**Offset**
0	Service Positive Response Identifier	62h		
1..2	DID SpeedInformation	FD00h		
3..4	Vehicle speed	xxxxh	0.05625km/h (Range: 0-296kph)	0
5..6	Wheel speed Front Left	xxxxh	0.05625km/h (Range: 0-296kph)	0
7..8	Wheel speed Front Right	xxxxh	0.05625km/h (Range: 0-296kph)	0
9..10	Wheel speed Rear Left	xxxxh	0.05625km/h (Range: 0-296kph)	0
11..12	Wheel speed Rear Right	xxxxh	0.05625km/h (Range: 0-296kph)	0

Remark: If there is failure at any of the wheel speed sensor, the corresponding vehicle speed/ wheel speed will be invalid and the output will be 0x7FFF

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6.5.3.6 *InputData*

The identifier reads information about several ECU inputs.

ECU → TESTER: [SERVICE] – Positive Response				
Byte	**Contents**	**Value + Unit**	**LSB**	**Comment**
0	Service Positive Response Identifier	62h		
1..2	DID InputData	FD01h		
3	Battery Voltage	xxh	0.08V	(0..20.4V)
4	Brake Light Switch	xxh		00h: OFF 01h: ON

6.5.3.7 *ActuationState*

The identifier allows to read information about the actuation state of the ECU's actuation.

ECU → TESTER: [SERVICE] – Positive Response			
Byte	**Bit**	**Contents**	**Value**
0		Service Positive Response Identifier	62h
1..2		DID ActuationState	FD02h
3		Valve Relay Status	00h = OFF, 01h= ON
4		Pump Motor Status	00h = OFF, 01h= ON
5	0	Valve Actuation State: Inlet Valve Front Left(EVFL)	0 = inactive, 1 = active
	1	Valve Actuation State: Outlet Valve Front Left(AVFL)	0 = inactive, 1 = active
	2	Valve Actuation State: Inlet Valve Front Right(EVFR)	0 = inactive, 1 = active
	3	Valve Actuation State: Outlet Valve Front Right(AVFR)	0 = inactive, 1 = active
	4	Valve Actuation State: Inlet Valve Rear Left(EVRL)	0 = inactive, 1 = active
	5	Valve Actuation State: Outlet Valve Rear Left(AVRL)	0 = inactive, 1 = active
	6	Valve Actuation State: Inlet Valve Rear Right(EVRR)	0 = inactive, 1 = active
	7	Valve Actuation State: Outlet Valve Rear Right(AVRR)	0 = inactive, 1 = active
6	0	Valve Actuation State: Valve USV1 (ESP Only)	0 = inactive, 1 = active
	1	Valve Actuation State: Valve USV2 (ESP Only)	0 = inactive, 1 = active
	2	Valve Actuation State: Valve HSV1 (ESP Only)	0 = inactive, 1 = active
	3	Valve Actuation State: Valve HSV2 (ESP Only)	0 = inactive, 1 = active
	4..7	Reserved	0000

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6.5.3.8 *FillingInStatus*

The identifier returns the FillingInStatus, which is stored in EEPROM.

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	62h
1	DID FillingInStatus	FD03h
2	FillingInStatus (hex, 1 bytes) • 00h : Filling-in not completed • AAh : Filling-in completed and ok • EEh : Filling-in completed and not ok	xxh

Remark: Bosch delivery state is FFh.

6.5.3.9 *EOLStatus*

The identifier returns the EOLStatus, which is stored in EEPROM.

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	62h
1	DID EOLStatus	FD04h
2	EOLStatus (hex, 1 bytes) • 00h: End-of-line test not completed • AAh: End-of-line test completed and ok • EEh: End-of-line test completed and not ok	xxh

Remark: Bosch delivery state is FFh.

6.5.3.10 *SystemSensor (ESP Only)*

The identifier reads sensors information from ECU.

ECU → TESTER: [SERVICE] – Positive Response				
Byte	**Contents**	**Value + Unit**	**LSB**	**Offset**
0	Service Positive Response Identifier	62h		
1..2	DID System Sensor	FD05h		
3..4	Master Cylinder Pressure	-42.5..425bar	0.0153bar	0
5..6	Steering Angle	-1440..1440deg	0.1deg	0
7..8	Yaw Rate	-2.1..2.1rad/s	0.00213rad/s	0
9..10	Lateral Acceleration	-21.6..21.6m/s^2	0.02712 m/s^2	0
11..12	Longitudinal Acceleration	-21.6..21.6m/ s^2	0.02712 m/s^2	0

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6.5.3.11 Variant Code (ESP Only)

The identifier reads Variant Code information from ECU.

ECU → TESTER: [SERVICE] – Positive Response			
Byte	**Contents**	**Value + Unit**	**Comment**
0	Service Positive Response Identifier	62h	
1..2	DID VariantCode	FD06h	
3	Variant Code	xxh	01h… 06h for Mainstream

6.5.4 Negative Response

ECU → TESTER: [SERVICE] – Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	22h
2	**Response Code** Incorrect Message Length Request Out Of Range Response Pending Conditions Not Correct General Reject	 13h 31h 78h 22h 10h

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6.6 CommunicationControl ($28h)

6.6.1 Description

The CommunicationControl service is used to switch on/off transmission and/or the reception of certain messages of ECU.

6.6.2 Request

PARAMETER: ControlType		
Size (Byte)	**Description**	**Value**
1 (Bit 0-6)	enableRxAndTx	00h
1 (Bit 0-6)	DisableRxAndTx	03h

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	28h
1	Sub Function Bit 0-6 ControlType Bit 7 Suppress Positive Response Message Indication Bit 0 = ECU is sending a positive response message 1 = ECU is not sending a positive response message	00/80h 03/83h
2	communication types: NormalCommunicationMessages	01h

6.6.3 Positive response

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	68h
1	SubFunction: ControlType	xxh

6.6.4 Negative Response

ECU → TESTER: [SERVICE] – Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	28h
2	**Response Code** Sub Function Not Supported Incorrect Message Length Request Out Of Range Session not support.	 12h 13h 31h 7fh

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6.7 SecurityAccess ($27h)

6.7.1 Description

The service SecurityAccess is used to get access to secured services by changing to different level of security. Changing the level of security is a two step procedure:

1. Tester requests seed, ECU sends seed
2. Tester sends the key, ECU sends a positive response (meaning the key was valid) and changes to the new security level

Remark:
The max number of false security key attempts is two. If a third wrong security key request is received the ECU is locked for 10 seconds and a NRC 36 (ExceededNumberOfAttempts) is sent.

PARAMETER: AccessModeRequestSeed		
Size (Byte)	**Description**	**Value**
1	Request Seed for General Security Level 1	01h

PARAMETER: AccessModeSendKey		
Size (Byte)	**Description**	**Value**
1	Send Key for General Security Level 1	02h

6.7.2 Request Step 1

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	27h
1	AccessModeRequestSeed	01h

6.7.3 Positive response Step 1

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	67h
1	AccessModeRequestSeed	01h
2	Seed byte 1 – High byte	xxh
3	Seed byte 2	xxh
4	Seed byte 3	xxh
5	Seed byte 4 – Low byte	xxh

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6.7.4 Request Step 2

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	27h
1	AccessModeSendKey	02h
2	Key byte 1 – High byte	xxh
3	Key byte 2	xxh
4	Key byte 3	xxh
5	Key byte 4 – Low byte	xxh

6.7.5 Positive response Step 2

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	67h
1	AccessModeSendKey	02h

6.7.6 Negative Response

ECU → TESTER: [SERVICE] – Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	27h
2	**Response Code** Sub Function Not Supported Incorrect Message Length Conditions Not Correct Request Sequence Error Invalid Key ExceededNumberOfAttempts RequiredTimeDelayNotExpired Session Not Support	 12h 13h 22h 24h 35h 36h 37h 7Fh

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6.8 WriteDataByIdentifier ($2Eh)

6.8.1 Description

The service allows the diagnostic tool to write data records into the ECU identified by Record Data Identifiers (DID). The client request message contains one 2-byte Data Identifier values that identifies the requested Data Record.

PARAMETER: Record Identifier		
Size (Byte)	**Description**	**Value**
2	Filling-in status of the hydraulic block	FD03h
2	EOLStatus	FD04h
2	Variant Code **(ESP Only)**	FD06h
2	Vehicle Identification Number -VIN	F190h

6.8.2 Request

6.8.2.1 Data Identifier = Filling-in status of the hydraulic block

This request is used to write a byte which indicates the filling-in status of the hydraulic block.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	2Eh
1..2	DID = [Filling-in status of the hydraulic block]	FD03h
3	Filling-in status: • 00h : Filling-in not completed • AAh : Filling-in completed and ok • EEh : Filling-in completed and not ok	xxh

Remark: Bosch delivery state is FFh.

6.8.2.2 Data Identifier = EOL Status

This request is used to write a byte which indicates the status if the end-of-line test.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	2Eh
1..2	DID = [EOL Status]	FD04h
3	End-of-line test status: • 00h : End-of-line test not completed • AAh : End-of-line test completed and ok • EEh : End-of-line test completed and not ok	xxh

Remark: Bosch delivery state is FFh.

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6.8.2.3 Data Identifier= Variant Code

This request is used to write the Variant Code information.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	2Eh
1..2	DID = [Variant Code]	FD06h
3	Variant Code	xxh

Remark: Supported variant code values are 01h…06h in Mainstream.

6.8.2.4 Data Identifier= Vehicle Identification Number -VIN

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	2Eh
1..2	DID = Vehicle Identification Number -VIN	F190h
3--19	VIN Code , 17 BYTEs	xxh

6.8.3 Positive response

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	6Eh
1..2	DID	xxxxh

6.8.4 Negative Response

ECU → TESTER: [SERVICE] - Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	2Eh
2	**Response Code** Incorrect Message Length Conditions Not Correct Request Out Of Range Security Access Denied Response Pending Session Not Support General Reject	 13h 22h 31h 33h 78h 7Fh 10h

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6.9 RoutineControl($31h)

6.9.1 Description

The service Routine Control is used by a diagnostic tool to start a routine, to stop a routine and to request the routine results.

PARAMETER: ControlOption		
Size (Byte)	**Description**	**Value**
1(0-6bit)	Start Routine	01h
1(0-6bit)	Stop Routine	02h
1(0-6bit)	Request routine results	03h
1(7bit)	0 = ECU is sending a positive response message 1 = ECU is not sending a positive response message	

Size (Byte)	Routine Identifier	Value
2	DelayedActuatorControl	F000h
2	EvacAndFill	F001h
2	RepairBleed	F002h
2	ActuatorTest	F003h
2	DynamicTest	F004h
2	WheelSpeedSensorsTest	F005h
2	Speed Limit Disable/Enable	F103h
2	Disable Valve Relay Shutdown	F104h
2	SAS Calibration **(Optional for ESP)**	F105h
2	YawRateSensorCalibration (ESP Only)	F106h

6.9.2 Requests

Remark: If the valves are activated for too long the ECU can be damaged. The maximum activation time of the valves must not exceed 30 seconds at an ambient temperature of 30°C. Using this service repeatedly can damage the valves or pump. Please keep this information in mind when using this service.

6.9.2.1 Emergency Stop

At any time it is possible to interrupt the control of the output which is being tested.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	31h
1	ControlOption = Stop Control	02h
2..3	Routine Identifier	xxxxh

Remark: If the ControlOption is set to “Stop control” the request must only have 4 bytes: Service Identifier, ControlOption, Routine Identifier.
Remark: Routine 0xF106, 0xF103, 0xF104 & 0xF105 cannot be stopped by ControlOption= Stop Control

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6.9.2.2 StartRoutine = DelayedActuatorControl

This request is used to control multiple drivers simultaneously. For each control two states are defined: state 1 and state 2. When the request was received the selected controls are set in state 1. After the activation time they are set to state 2.

In comparison to the Actuator Test this service allows the tester to actuate one or more valves for a specified time. This feature allows the tester to run the customer specified End Of Line routine sequence instead of Pre-defined Periodic Actuator Control (EvacAndFill, RepairBleed) sequence.

PARAMETER: ValveStates	
Position (Bit)	**Description**
0	Inlet front left
1	Outlet front left
2	Inleft front right
3	Outlet front right
4	Inlet rear left
5	Outlet rear left
6	Inlet rear right
7	Outlet rear right

PARAMETER: DriverStates	
Position (Bit)	**Description**
0	USV1 (ESP Only)
1	USV2 (ESP Only)
2	HSV1 (ESP Only)
3	HSV2 (ESP Only)
4	Unused
5	Unused
6	Pump
7	Unused

Remark: For each control the corresponding bit has to be set to 1 to switch the actuator on and to 0 to switch the actuator off.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	31h
1	ControlOption	01h
2..3	Routine Identifier = [DelayedActuatorControl]	F000h
4	ValveStates (State 1)	xxh
5	DriverStates (State 1)	xxh
6	Activation time (1LSB = 20ms)	xxh
7	ValveStates (State 2)	xxh
8	DriverStates (State 2)	xxh

Remark: A “Failure” response has one of the following reasons: At least one valve or one driver doesn’t exist or the control is not possible at this time because of safety reasons.

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Remark: A control that has been started by a delay actuator control request can be stopped by a request with the ControlOption set to “Stop control” (Emergency Stop, 31h 02h F0h 00h).

Remark: At each end of this service the tester has to send the following request to turn off all controls: 31h 01h F0h 00h 00h 00h 00h 00h 00h.

6.9.2.3 StartRoutine = EvacAndFill

This request is used to fill and drain the hydraulic unit.

TESTER → ECU: [SERVICE] – Request				
Byte	**Contents**	**Value + Unit**	**LSB**	**Offset**
0	Service Identifier	31h		
1	ControlOption	01h		
2..3	Routine Identifier = [EvacAndFill]	F001h		
4	Evac and Fill time	xxh (2..150s)	1s	0
5	Time until start of pump motor	Xxh (2..150s) FF = no pump motor	1s	0

Remark: Time has to be between 2 s and 150 s. If Time exceeds this limit, the ECU uses the maximum defined value.
After evac and filling time, the pump motor will run for 2 additional seconds, if the delay time in byte 5 is not FF.

6.9.2.4 StartRoutine = RepairBleed

TESTER → ECU: [SERVICE] – Request			
Byte	**Contents**	**Value + Unit**	**Comment**
0	Service Identifier	31h	
1	Control Option	01h	
2..3	Routine Identifier = [RepairBleed]	F002h	
4	Repair Bleeding phase	0x01 = phase1 0x02 = phase2 0x03 = phase3 0x04 = phase4	phase 1..4
5	Repetition counter	0x03-0x05	No. of loops: 3..5

Remark:
<u>Repair bleeding phase</u>
The repair bleeding function should always be executed in increasing phase sequence starting with Phase 1. A repetition of the same phase is allowed, but no phase shall be skipped. If this is not ensured, the pump might be damaged due to dry run.
<u>Repetition counter</u>
This parameter defines the number of inner loop repetitions in phase 3 and 4. This parameter is always required, but only valid for Phase 3 and 4, for other phases, it is ignored.

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6.9.2.5 *StartRoutine = ActuatorTest*

This request is used to start Actuator Test. Actuator table refer to Chapter 7.Actuator Table. Actuator 1,2 immediately start actuation . After Actuation time Actuator 3,4 start actuation.

TESTER → ECU: [SERVICE] – Request				
Byte	**Contents**	**Value + Unit**	**LSB**	**Offset**
0	Service Identifier	31h		
1	ControlOption	01h		
2..3	Routine Identifier = [ActuatorTest]	F003h		
4..5	Actuator 1 Value	xxxxh		
6..7	Actuator 1 Identifier	xxxxh		
8..9	Actuator 2 Value	xxxxh		
10..11	Actuator 2 Identifier	xxxxh		
12..13	Actuation time	10ms..327s	10ms	0
14..15	Actuator 3 Value	xxxxh		
16..17	Actuator 3 Identifier	xxxxh		
18..19	Actuator 4 Value	xxxxh		
20..21	Actuator 4 Identifier	xxxxh		

Remark: The pump motor relay must not be actuated for times shorter than 20ms. Shorter actuations could damage the motor relay due to very high current during the starting phase of the motor.

Remark: Bosch strongly recommends only using an ActuationValue of 0% and 100%.

Remark: A negative response has one of the following reasons: At least one valve or one driver doesn't exist or the control is not possible at this time because of safety reasons.

Remark: If the valve relay is out of order, the routine will be aborted.

6.9.2.6 *StartRoutine = DynamicTest*

This request is used to start Dynamic Test.

TESTER → ECU: [SERVICE] – Request				
Byte	**Contents**	**Value + Unit**	**LSB**	**Offset**
0	Service Identifier	31h		
1	ControlOption	01h		
2..3	Routine Identifier = [DynamicTest]	F004h		
4..5	Actuator 1 Value	xxxxh		
6..7	Actuator 1 Identifier	xxxxh		
8..9	Actuator 2 Value	xxxxh		
10..11	Actuator 2 Identifier	xxxxh		
12..13	Actuation time	10ms..327s	10ms	0
14..15	Actuator 3 Value	xxxxh		
16..17	Actuator 3 Identifier	xxxxh		
18..19	Actuator 4 Value	xxxxh		
20..21	Actuator 4 Identifier	xxxxh		
22..23	Delay Time	10ms..327s	10ms	0

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6.9.2.7 StartRoutine = WheelSpeedSensorsTest

This request is used to start Wheel Speed Sensors Test.

TESTER → ECU: [SERVICE] – Request				
Byte	**Contents**	**Value + Unit**	**LSB**	**Offset**
0	Service Identifier	31h		
1	ControlOption	01h		
2..3	Routine Identifier = [WheelSpeedSensorsTest]	F005h		
4..5	Execution time	xxxxh	10ms	0

Remark: If execution time is 0, then NRC13 will be responded.

6.9.2.8 StartRoutine = Speed Limit Disable/Enable

This request is Disable/ Enable the speed Limit (10kph for Extended Mode)

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value + Unit**
0	Service Identifier	31h
1	ControlOption	01h
2..3	Routine Identifier = [Speed Limit Disable/Enable]	F103h
4	Control Option 00h: Disable speed Limit 01h: Enable Speed Limit	xxh

6.9.2.9 StartRoutine = Disable Valve Relay Shutdown

This request is used to disable/ enable Valve Relay shutdown when one of the valves is faulty.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value + Unit**
0	Service Identifier	31h
1	ControlOption	01h
2..3	Routine Identifier = [Disable Valve Relay shutdown]	F104h
4	Control Option 00h: Disable Valve Relay Shutdown 01h: Re-enable Valve relay shutdown	xxh

6.9.2.10 StartRoutine = SAS Calibration

This request is used to calibrate the SAS (Steering Angle Sensor)

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value + Unit**
0	Service Identifier	31h
1	ControlOption	01h
2..3	Routine Identifier = [SAS Calibration]	F105h

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6.9.2.11 *StartRoutine = YawRateSensorCalibration*

This request is used to start Yaw Rate Sensor calibration.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value + Unit**
0	Service Identifier	31h
1	ControlOption	01h
2..3	Routine Identifier = [YawRateSensorCalibration]	F106h

6.9.2.12 *Request Routine Results*

This request is used to gets the results of routines previously activated with the Start Routine.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	31h
1	ControlOption = Request routine results	03h
2..3	Routine Identifier	xxxxh

6.9.3 Supported Control/Routine Status in Positive response

PARAMETER: supported Control/Routine Status	
Size (Byte)	**Description**
4	Supported Control Status in Positive Response of Sub Function == Start Routine 01 = start successful 02 = already started 03 = not possible to start
4	Supported Control Status in Positive Response of Sub Function == Stop Routine 01 = stop successful 02 = already stopped or not start 03 = routine stopped abnormal
4	Supported Control Status in Positive Response of Sub Function == Request Routine Results 01 = finished 02 = still running 03 = stopped or not start 04 = routine stopped abnormal

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6.9.3.1 *StartRoutine and StopRoutine*

For requests with Control option Start Routine (01h) and Stop Routine (02h), the positive response is returned in the following format.

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	71h
1	ControlOption (01h or 02h)	Xxh
2..3	Routine Identifier	xxxxh
4	Supported Control Status	Xxh

6.9.3.2 *RequestRoutineResult = DelayedActuatorControl, EvacAndFill, RepairBleed, SASCalibration,YawRateSensorCalibration.*

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	71h
1	ControlOption	03h
2..3	Routine Identifier	Xxxxh
4	Supported Control Status	xxh

6.9.3.3 *RequestRoutineResult = ActuatorTest*

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	71h
1	ControlOption	03h
2..3	Routine Identifier	F003h
4	Supported Control Status	xxh
5..6	Result, actuator 1	Xxxxh
7..8	Result, actuator 2	Xxxxh
9..10	Result, actuator 3	Xxxxh
11..12	Result, actuator 4	Xxxxh

Remark: Bytes 5…12 are only sent if the response is “RoutineStatus = Routine finished”.

6.9.3.4 *RequestRoutineResult = DynamicTest*

Byte	Contents	Value + Unit	LSB
0	Service Positive Response Identifier	71h	
1	ControlOption	03h	
2..3	Routine Identifier	F004h	
4	Supported Control Status	xxh	
5..6	delta wheel speed front left	-100..100m/s (0xe700 -- 0x1900) m/s	1/64 m/s
7..8	delta wheel speed front right	-100..100m/s (0xe700 -- 0x1900) m/s	1/64 m/s
9..10	delta wheel speed rear left	-100..100m/s	1/64 m/s

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		(0xe700 -- 0x1900) m/s	
11..12	delta wheel speed rear right	-100..100m/s (0xe700 -- 0x1900) m/s	1/64 m/s

Remark: Bytes 5…12 are only sent if the response is “RoutineStatus = Routine finished”.

6.9.3.5 RequestRoutineResult = WheelSpeedSensorsTest

Byte	Contents	Value + Unit	LSB
0	Service Positive Response Identifier	71h	
1	ControlOption	03h	
2..3	Routine Identifier	F005h	
4	Supported Control Status	Xxh	
5..6	Maximum speed of the front left wheel	0 m/s-- 100m/s (Hex 0x0000 – 0x1900) m/s	1/64 m/s
7..8	Minimum speed of the front left wheel	0 m/s-- 100m/s (Hex 0x0000 – 0x1900) m/s	1/64 m/s
9..10	Maximum speed of the front right wheel	0 m/s-- 100m/s (Hex 0x0000 – 0x1900) m/s	1/64 m/s
11..12	Minimum speed of the front right wheel	0 m/s-- 100m/s (Hex 0x0000 – 0x1900) m/s	1/64 m/s
13..14	Maximum speed of the rear left wheel	0 m/s-- 100m/s (Hex 0x0000 – 0x1900) m/s	1/64 m/s
15..16	Minimum speed of the rear left wheel	0 m/s-- 100m/s (Hex 0x0000 – 0x1900) m/s	1/64 m/s
17..18	Maximum speed of the rear right wheel	0 m/s-- 100m/s (Hex 0x0000 – 0x1900) m/s	1/64 m/s
19..20	Minimum speed of the rear right wheel	0 m/s-- 100m/s (Hex 0x0000 – 0x1900) m/s	1/64 m/s

Remark: Bytes 5…20 are only sent if the response is “RoutineStatus = Routine finished”.
Remark: If any of the wheel speed sensor is invalid, the output will be 0xFFFF
Remark: Routine ID F103h, F104h are not supported by Request RoutineResult

6.9.4 Negative Response

ECU → TESTER: [SERVICE] – Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	31h
2	**Response Code** Sub Function Not Supported Incorrect Message Length Or Invalid Format Conditions Not Correct Request Out Of Range Security Access Denied Session Not Support	 12h 13h 22h 31h 33h 7Fh

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6.10 TesterPresent ($3Eh)

6.10.1 Description

This service is used to indicate to an ECU that the diagnostic tool is still connected to the vehicle and that diagnostic sessions are to remain active.

6.10.2 Request

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	3Eh
1	Sub Function Bit 0-6 00 Bit 7 Suppress Positive Response Message Indication Bit 0 = ECU is sending a positive response message 1 = ECU is not sending a positive response message	00/80h

6.10.3 Positive response

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	7Eh
1	SubFunction	xxh

6.10.4 Negative Response

ECU → TESTER: [SERVICE] – Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	3Eh
2	**Response Code** Sub Function Not Supported Incorrect Message Length	 12h 13h

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6.11 ControlDTCSetting ($85h)

6.11.1 Description

This service is used to disable or enable the storing of Diagnostic Trouble Codes.

6.11.2 Request

PARAMETER: DTCSettingType		
Size (Byte)	**Description**	**Value**
1 (Bit 0-6)	The ECU resumes the setting of diagnostic trouble codes according to normal operating conditions.	01h
1 (Bit 0-6)	The ECU stops the setting of diagnostic trouble codes.	02h

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	85h
1	Sub Function Bit 0-6 DTCSettingType Bit 7 Suppress Positive Response Message Indication Bit 0 = ECU is sending a positive response message 1 = ECU is not sending a positive response message	01/81h 02/82h

6.11.3 Positive response

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	C5h
1	SubFunction	xxh

6.11.4 Negative Response

ECU → TESTER: [SERVICE] – Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	85h
2	**Response Code** Sub Function Not Supported Incorrect Message Length Session Not Support	 12h 13h 7Fh

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7 Actuator Table

The following table lists the available actuator identifier for service $31. In particular, they are relevant for the actuator test and dynamic test.

PARAMETER: Actuator Number		
Actuator-ID	**Channel**	**Actuator-Value**
0000h	No actuation	0xFFFF=ON 0x0000=OFF
0022h	Pump motor relay	0xFFFF=ON 0x0000=OFF
0030h	Inlet valve front left	0xFFFF=ON 0x0000=OFF
0032h	Outlet valve front left	0xFFFF=ON 0x0000=OFF
0034h	Inlet valve front right	0xFFFF=ON 0x0000=OFF
0036h	Outlet valve front right	0xFFFF=ON 0x0000=OFF
0038h	Inlet valve rear right	0xFFFF=ON 0x0000=OFF
003Ah	Outlet valve rear right	0xFFFF=ON 0x0000=OFF
003Ch	Inlet valve rear left	0xFFFF=ON 0x0000=OFF
003Eh	Outlet valve rear left	0xFFFF=ON 0x0000=OFF
004Eh	USV1 **(ESP Only)**	0xFFFF=ON 0x0000=OFF
0050h	USV2 **(ESP Only)**	0xFFFF=ON 0x0000=OFF
0052h	HSV1 **(ESP Only)**	0xFFFF=ON 0x0000=OFF
0054h	HSV2 **(ESP Only)**	0xFFFF=ON 0x0000=OFF
0002h	ABS Safety Lamp	0xFFFF=ON 0x0000=OFF
0004h	ESP Safety Lamp**(ESP Only)**	0xFFFF=ON 0x0000=OFF
000Eh	EBD Safety Lamp	0xFFFF=ON 0x0000=OFF

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8 List of DTCs

8.1 DTC List

DTC	DTC (Hex)	DTC Description	ABS /ESP	Notice
C190004	**0x590004**	ECU Voltage supply: high voltage	A/E	
C190104	**0x590104**	ECU Voltage supply: low voltage	A/E	
C100004	**0x502004**	ECU error (hardware, Micro-controller error)	A/E	
C101008	**0x503008**	ECU error (general software problem)	A/E	
C006B06	**0x406B06**	ABS/ESP implausible control (control time too long, etc.)	A/E	
C003108	**0x403108**	Wheel-speed sensor, front left: (signal failure) out of range, lost, noise, Intermittent	A/E	
C003200	**0x403200**	Wheel-speed sensor, front left: Signal Line short to GND/Open; Supply Line open	A/E	
C00A000	**0x40A000**	Wheel-speed sensor, front left: Supply Line short to GND	A/E	
C00A100	**0x40A100**	Wheel-speed sensor, front left: Signal Line short to UBATT	A/E	
C00A900	**0x40A900**	Wheel-speed sensor, front left: general error	A/E	undef (no exact failure reason detectable)
C003408	**0x403408**	Wheel-speed sensor, front right: (signal failure) out of range, lost, noise, Intermittent	A/E	
C003500	**0x403500**	Wheel-speed sensor, front right: Signal Line short to GND/Open; Supply Line open	A/E	
C00A200	**0x40A200**	Wheel-speed sensor, front right: Supply Line short to GND	A/E	
C00A300	**0x40A300**	Wheel-speed sensor, front right: Signal Line short to UBATT	A/E	

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DTC	DTC (Hex)	DTC Description	ABS /ESP	Notice
C00AA00	**0x40AA00**	Wheel-speed sensor, front right: general error	A/E	undef (no exact failure reason detectable)
C003708	**0x403708**	Wheel-speed sensor, rear left: (signal failure)out of range, lost, noise, Intermittent	A/E	
C003800	**0x403800**	Wheel-speed sensor, rear left: Signal Line short to GND/Open; Supply Line open	A/E	
C00A400	**0x40A400**	Wheel-speed sensor, rear left: Supply Line short to GND	A/E	
C00A500	**0x40A500**	Wheel-speed sensor, rear left: Signal Line short to UBATT	A/E	
C00AB00	**0x40AB00**	Wheel-speed sensor, rear left: general error	A/E	undef (no exact failure reason detectable)
C003A08	**0x403A08**	Wheel-speed sensor, rear right: (signal failure) out of range, lost, noise, Intermittent	A/E	
C003B00	**0x403B00**	Wheel-speed sensor, rear right: Signal Line short to GND/Open; Supply Line open	A/E	
C00A600	**0x40A600**	Wheel-speed sensor, rear right: Supply Line short to GND	A/E	
C00A700	**0x40A700**	Wheel-speed sensor, rear right: Signal Line short to UBATT	A/E	
C00AC00	**0x40AC00**	Wheel-speed sensor, rear right: General error	A/E	undef (no exact failure reason detectable)
C109904	**0x509904**	Wheel-speed sensor generic error (sensor swap, multiple sensors error)	A/E	
C004460	**0x404460**	Pressure sensor fault (Signal)	E	
C004510	**0x404510**	Pressure sensor fault (Line)	E	
C004008	**0x404008**	BLS fault: Plausibility	E	

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DTC	DTC (Hex)	DTC Description	ABS /ESP	Notice
U160108	**0xD60108**	BLS Net signal invalid value (via CAN)	E	
C006108	**0x406108**	Lateral acceleration sensor: signal fault	E	
C006208	**0x406208**	Longitudinal acceleration sensor: signal fault	E	
C006308	**0x406308**	Yaw Rate Sensor: signal fault	E	
C019604	**0x419604**	YRS sensor cluster fault (Hardware failure, Temperature , range, internal fault)	E	
C00A800	**0x40A800**	YRS sensor cluster not calibrated or calibration failed	E	
U000500	**0xC00500**	CAN Bus overvoltage	A/E	
U000700	**0xC00700**	CAN Bus under voltage	A/E	
C100104	**0x500104**	CAN hardware error	A/E	
U000104	**0xC00104**	CAN Bus Off error	A/E	
U100104	**0xD00104**	CAN Error Passive	A/E	
U010004	**0xC10004**	CAN timeout EMS	E	
U010008	**0xC10008**	CAN data corrupt / signal invalid EMS	E	
C106600	**0x506600**	SAS Calibration Error	E	Optional: ESP projects only, recommended
C046008	**0x446008**	SAS fault (Signal)	E	From ASW, SAS signal layer algorithm/logic error

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DTC	DTC (Hex)	DTC Description	ABS /ESP	Notice
U012604	**0xC12604**	CAN timeout SAS	E	
U012608	**0xC12608**	CAN data corrupt SAS (DLC, CRC, alive counter error, etc)	E	
U010104	**0xC10104**	CAN timeout TCU	E	
U010108	**0xC10108**	CAN data corrupt/signal invalid TCU	E	
C001004	**0x401004**	valve fault, inlet valve FL	A/E	
C001104	**0x401104**	valve fault, outlet valve FL	A/E	
C001404	**0x401404**	valve fault, inlet valve FR	A/E	
C001504	**0x401504**	valve fault, outlet valve FR	A/E	
C001804	**0x401804**	valve fault, inlet valve RL	A/E	
C001904	**0x401904**	valve fault, outlet valve RL	A/E	
C001C04	**0x401C04**	valve fault, inlet valve RR	A/E	
C001D04	**0x401D04**	valve fault, outlet valve RR	A/E	
C000104	**0x400104**	Valve fault, USV1	E	

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DTC	DTC (Hex)	DTC Description	ABS /ESP	Notice
C000204	**0x400204**	Valve fault, USV2	E	
C000304	**0x400304**	Valve fault, HSV1	E	
C000404	**0x400404**	Valve fault, HSV2	E	
C109504	**0x509504**	Valve relay error	A/E	
C002004	**0x402004**	Return pump fault	A/E	
C007208	**0x407208**	General Valve fault (overheat protection, signal invalid, hardware fault)	A/E	
C104C04	**0x504C04**	PATA switch failure	E	
C121208	**0x521208**	Variant Coding Error	E	Optional
U160208	**0xD60208**	Handbrake signal invalid value (via CAN)	E	Optional: only applicable if handbrake signal sent via CAN
C100300	**0x500300**	Clutch switch error	E	Clutch signal is permanently on high/low level
U160308	**0xD60308**	Clutch signal invalid value (via CAN)	E	Optional: only applicable if clutch signal sent via CAN
U160408	**0xD60408**	Reverse Gear Switch signal invalid value (via CAN)	E	Optional: HHC project only; only applicable if Reverse Gear Switch signal sent via CAN

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DTC	DTC (Hex)	DTC Description	ABS /ESP	Notice
C108C08	**0x508C08**	Reverse Gear Switch signal permanent High/Low	E	Optional: HHC project only; only applicable if Reverse Gear Switch signal is sent
U012308	**0xC12308**	Lost Communication With Yaw Rate Sensor Module	E	Optional: External YRS sensor cluster used.
U051308	**0xC51308**	Invalid Data Received From Yaw Rate Sensor Module	E	Optional: External YRS sensor cluster used.
C006102	**0x406102**	External YRS sensor cluster of Lateral acceleration sensor: signal fault	E	Optional: External YRS sensor cluster used.
C006202	**0x406202**	External YRS sensor cluster of Longitudinal acceleration sensor: signal fault	E	Optional: External YRS sensor cluster used.
C006302	**0x406302**	External YRS sensor cluster of Yaw Rate Sensor: signal fault	E	Optional: External YRS sensor cluster used.
U014004	**0xC14004**	CAN timeout BCM	E	Optional: Only BCM is node is available.
U014008	**0xC14008**	CAN data corrupt BCM (DLC, CRC, alive counter error, etc)	E	Optional: Only BCM is node is available.
U014604	**0xC14604**	CAN timeout Gateway	E	Optional: Only Gateway node is available.
U014608	**0xC14608**	CAN data corrupt Gateway (DLC, CRC, alive counter error, etc)	E	Optional: Only Gateway node is available.

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